

Capstone Project Proposal

AI-Powered Financial Operating System

For Startups and SMEs

Submitted under the guidance of

Professor Jinraj Joshipura

Submitted by

Parin Patel	B.Tech CSE
Kavish Shah	B.Tech CSE
Heer Gandhi	B.Tech CSE
Nirmam Parikh	B.Tech CSE
Dhruv Patel	BBA Finance
Nisarg Patel	BBA Marketing

Ahmedabad University
Capstone Project

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1 Project Title

AI-Powered Financial Operating System for Startups and SMEs

2 Project Overview

This project proposes the development of a financial operating system for startups and small and medium-sized enterprises that transforms raw business data into actionable financial intelligence.

The platform is designed to act as an AI-powered virtual CFO for founders and business owners who may not have deep financial expertise. It will help users understand their financial position, track key financial metrics, simulate business decisions, and generate reports useful for internal management, board discussions, and investor readiness.

3 Problem Statement

Many startup founders and small business owners understand their products, customers, and operations well, but they are not necessarily finance experts. As a result, they often struggle to answer important financial questions such as:

- Can we afford to hire more employees?
- How long can we survive without additional funding?
- Which expenses are hurting profitability?
- What happens if revenue drops?
- Are we financially ready to approach investors?

Most small businesses do not maintain sophisticated financial statements. Hiring a full-time Chief Financial Officer is often expensive and not feasible for early-stage startups or SMEs. This creates a gap between business data and financial decision-making.

4 Proposed Solution

The proposed platform acts as an AI-powered virtual CFO. Instead of requiring users to have strong financial knowledge, the system allows them to either upload existing financial documents or enter raw business data manually.

The platform then automatically processes this information and converts it into meaningful financial insights, KPIs, simulations, dashboards, and reports.

5 Data Input Options

5.1 Option 1: Upload Existing Financial Documents

For users who already maintain financial records, the platform will allow uploading of:

- Income Statements
- Cash Flow Statements
- Balance Sheets
- Accounting Exports
- Excel or CSV Files

5.2 Option 2: Enter Raw Business Data

For early-stage founders and small businesses that do not maintain formal financial statements, the platform will allow manual entry of raw business data such as:

- Sales
- Purchases
- Salaries
- Rent
- Marketing Expenses
- Operational Costs
- Bank Transactions

6 Core Workflow

6.1 Step 1: Data Collection

Users upload documents or manually enter business data into the platform. This data becomes the foundation for financial analysis and reporting.

6.2 Step 2: Financial Intelligence Engine

The platform automatically processes the collected data by performing the following functions:

- Categorizing transactions
- Calculating revenue and expenses
- Estimating cash flow
- Generating financial summaries
- Tracking historical performance

6.3 Step 3: KPI Generation

The system calculates important startup and SME financial metrics, including:

- Burn Rate
- Runway
- Gross Margin
- Operating Margin
- Revenue Growth
- Profitability Trends

6.4 Step 4: Scenario Simulation

Users can test business decisions before making them. For example, the platform can simulate the impact of:

- Hiring three employees
- Increasing the marketing budget by 50%
- Raising product prices by 10%
- Experiencing a 20% revenue decline

The system immediately shows the impact of these scenarios on:

- Cash flow
- Runway
- Profitability
- Growth

6.5 Step 5: AI CFO Assistant

The AI CFO Assistant interprets the financial data and answers practical business questions such as:

- Can we afford expansion?
- Why are margins declining?
- What expenses should we optimize?
- Should we prioritize growth or cost reduction?

The assistant explains financial concepts in simple language suitable for founders and non-finance users.

6.6 Step 6: Automated Reports

The platform generates structured reports such as:

- Monthly Financial Reports
- Board Reports
- Investor Readiness Summaries
- Financial Health Snapshots

7 Differentiation from ChatGPT

ChatGPT is a general-purpose AI tool. In contrast, the proposed platform is designed specifically for financial decision support for startups and SMEs.

The platform will:

- Maintain company financial history
- Track KPIs automatically
- Generate dashboards
- Run structured business simulations
- Produce financial reports
- Provide startup-specific financial workflows

The AI component is only one part of the system. The real value lies in continuously transforming business data into structured financial intelligence and decision support.

8 Expected Outcome

The expected outcome of this capstone project is a working prototype of an AI-powered financial operating system that enables startups and SMEs to:

- Understand their financial health
- Track important KPIs
- Make informed business decisions
- Simulate financial scenarios
- Generate investor and board-ready reports
- Reduce dependency on expensive financial consultants at an early stage

9 Team Members

Name	Program
Parin Patel	B.Tech CSE
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10 Conclusion

This project aims to bridge the gap between raw business data and financial decision-making for startups and SMEs. By combining automated financial processing, KPI tracking, scenario simulation, AI-based interpretation, and structured reporting, the platform can serve as a practical virtual CFO for businesses that cannot afford dedicated financial leadership.